

Jared Carreno

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EDUCATION

Harvey Mudd College

B.S., Engineering, GPA: 3.3/4.0

Claremont, CA

May 2026

TECHNICAL SKILLS

Electronics & Firmware: KiCAD, SystemVerilog, Radiant Lattice, Quartus Prime, QuestaSim, LTspice

Software & Scripting: Python, C, C++, MATLAB, Java, LabVIEW

Mechanical Design: SolidWorks (FEA, CAD), PTC Creo, COMSOL, Machining, Product Design and Development

RELEVANT COURSEWORK

Analog Circuit Design • Continuum Mechanics • Manufacturing and Design • Thermodynamics • Materials Engineering
Signals & Systems • Control Systems • Digital Signal Processing • State Estimation • Microprocessor System Design

EXPERIENCE

Electromechanical Engineer (Industry Capstone)

August 2025 – May 2026

Ohmium

Claremont, CA

- Redesigned an existing hydrogen electrolyzer cell to perform concurrent, independent electrolysis on 4 distinct materials, increasing experimental throughput by 400%.
- Led all mechanical design in SolidWorks, engineering a custom anode insulator and insulating cross designed for seamless flow channel integration.
- Fabricated the assembly via manual milling and 3D printing, executing rigorous root cause analysis to systematically eliminate fluid leaks and ensure continuous system integrity.
- Designed a high-current mixed-signal PCB in KiCAD for independent current regulation and live data collection.

Mechanical Engineering Intern

May 2025 – August 2025

HARMAN International (Samsung)

Northridge, CA

- Engineered custom, modular acoustic enclosures for smart home and consumer sleep applications that directly helped secure corporate partnerships.
- Partnered with a team to retrofit a commercial child car seat with an embedded audio system, engineering sound-permeable TPU cushions in SolidWorks and tuning the system's frequency response using SigmaStudio.
- Validated hardware reliability across commercial and consumer lines, performing ultimate tensile testing on stadium speaker trusses and in-house IPX water resistance tests.

Electromechanical Engineer (Industry Capstone)

Jan 2025 – May 2025

Trilobio

Claremont, CA

- Worked in a team to develop an electromechanical analytical balance designed to integrate into an existing robotic system and automate the weight validation process.
- Modeled the entire system in SolidWorks and Onshape; machined housing and flexures with a 3D-printed damping mechanism.
- Engineered sensitive, custom mixed-signal PCBs in KiCAD to relate mass to capacitance data.
- Tested, troubleshot, and validated system performance to reduce noise, ensure accuracy, and facilitate integration.

PROJECTS

DIY Operational Amplifier

May 2026

Harvey Mudd College

Claremont, CA

- Architected and built a stable operational amplifier from scratch using BJTs, integrating a balun, differential amplifier, gain stage, and Class AB output stage.
- Optimized the design to minimize interstage loading and stabilize DC operating points, validating final performance via LTspice and benchtop instrumentation.

IoT Temperature Sensor

December 2025

Harvey Mudd College

Claremont, CA

- Developed a wireless communication bridge using an ESP8266 module to enable remote data transmission from an STM32 over Wi-Fi.
- Implemented dual-interface communication leveraging SPI for sensor readings and UART for ESP8266 and MCU communication.